

Calibration Recipes for Anonymous DHT Deployments: Budgets, Horizons, and Alarm Tax

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Abstract

To keep the main Anonymous DHT papers referee-tight and non-redundant, we collect the “numbers-first” calibration tables and rule-of-thumb conversions that otherwise get repeated: (i) witness-amplification horizons for state-dependent leakage, (ii) translating observation cadence into time-to-detection, (iii) the inevitable alert-oracle tax for closed-view auditing. We also include pointers to the CPPC manifest interface (AnonDHT State 3) without reprinting schemas or cards.[2] Core definitions and theorems live in the State series and Synthesis notes; this paper is purely an operational bridge.

Scope. This is an operational bridge: “numbers-first” recipes and tables that help size witness cadences, horizons, and alarm costs without repeating algebra across the core papers.

Units. Budgets are published in bits (use $\log_2$ and 2^ϵ); results stated in nats can be converted by dividing by $\ln 2$ (see Synthesis 8).[9]

Imports. We import the threat/window checklist and trace-interface vocabulary.[5, 4] Amplification bounds are imported from the State series (Papers 1 and 3), and identification sizing rules from the endpoint bridge note.[1, 2, 3]

Exports. We export compact calibrations: (i) quadratic horizon scaling for persistent witness gaps, (ii) cadence-to-horizon translations, and (iii) a tabulated alert-oracle “tax” lower bound for closed-view auditing. For the CPPC manifest interface, see AnonDHT State 3.[2]

Artifact policy. This archive is source-only; the full CPPC schema and supporting tooling are available on request.[8]

1 Series context and what this note owns

We use the threat/window checklist [5] and the trace-interface vocabulary [4]. This note owns only “back-of-the-envelope” translations that help readers size cadence, windows, and budgets without re-deriving the same algebra in every paper.

2 Recipe 1: witness amplification needs quadratic horizons

State series Paper 1 [1] shows that even a tiny, persistent witness bias amplifies under repetition. A convenient calibration is the Hoeffding-style rule of thumb

$$T \approx \frac{2\log_2(1/\eta)}{\Delta^2},$$

where Δ is the per-epoch witness gap and η is a target total error probability.

Per-epoch gap Δ	Target total error η	Rule-of-thumb T	
0.01	0.05	$\approx 6 \times 10^4$	tiny bias needs weeks at minute cadence (or days at 5s)
0.02	0.05	$\approx 1.5 \times 10^4$	~ 10 days at minute cadence (or ~ 2 months at 5s)
0.05	0.05	$\approx 2.4 \times 10^3$	~ 40 minutes at 1s cadence (or ~ 7 hours at 5s)
0.10	0.05	$\approx 6 \times 10^2$	~ 10 minutes at 1s cadence (or ~ 1.2 hours at 5s)

Table 1: Calibration for witness amplification (Paper 1 [1]). Epoch vocabularies vary across systems (e.g. Nym uses minute-scale epochs [6]); the key scaling is the quadratic dependence on Δ .

Takeaway. *If an adversary can extract even a 1–2% per-epoch witness gap and the system runs at high cadence, “state” becomes a fast privacy amplifier unless it is refreshed/privatized.*

3 Recipe 2: budget-to-horizon translation (cadence $\rightarrow$ time)

If a monitoring/evidence stream yields T independent-ish epochs per window, the same T can correspond to very different wall-clock horizons depending on cadence and sampling. Table 2 gives an illustrative translation assuming the rule of thumb above with $\eta = 0.05$.

Per-epoch gap Δ	T (epochs)	Horizon at $10^3/\text{day}$	Horizon at $10^5/\text{day}$
0.01	6×10^4	≈ 60 days	≈ 14 hours
0.02	1.5×10^4	≈ 15 days	≈ 3.6 hours
0.05	2.4×10^3	≈ 2.4 days	≈ 35 minutes

Table 2: Budget-to-horizon calibration for witness amplification (Paper 1 [1]). If an observer sees only a fraction q of epochs, multiply horizons by $1/q$. If state is refreshed every W days, amplification effectively saturates once T exceeds the refresh horizon.

4 Recipe 3: the alert-oracle tax is logarithmic but not free

Closed-view auditing (State series Paper 3 [2]) can require “alarms” or public alerts. If an alert has miss probability β and false-alarm probability α , the paper’s alert-oracle bound yields a necessary privacy cost roughly

$$\varepsilon \gtrsim \log_2\left(\frac{1-\beta}{\alpha}\right),$$

independent of mechanism details.

Miss prob. β	False-alarm α	Lower bound $\varepsilon \gtrsim \log_2((1 - \beta)/\alpha)$
0.10	0.05	2.89
0.10	0.01	4.50
0.10	0.001	6.80
0.05	0.05	2.94
0.05	0.01	4.55
0.05	0.001	6.86
0.01	0.05	2.99
0.01	0.01	4.60
0.01	0.001	6.90

Table 3: Calibration for the alert-oracle tax (Paper 3 [2]). Tight alarms cost nontrivial ε ; pushing α down by 1000x costs about 6.9 nats $\approx$ 10 bits.

Takeaway. *Alarm quality is a privacy knob: to get rare false alarms you must usually spend a meaningful privacy budget, even before any “payload” disclosures.*

5 Recipe 4: endpoint sizing via MaxL and odds inflation (pointer)

For identification and shortlisting conversions (e.g. “what MaxL budget prevents exact identification among K candidates?”), we do not restate algebra here. Use the endpoint bridge note [3], which relates realized odds inflation, pointwise maximal leakage, and maximal leakage and records the standard $\log_2(\alpha K)$ sizing rule.

For destination-privacy equalization of routing transcripts, MC-EQ/CoverSketch provides an explicit session scaling bound under uniform covers: $\mathcal{L}_{\max} \leq T \log_2(1 + 2k\delta_{eq})$ (receipt-friendly and conservative).[7]

6 CPPC schema pointer

For the CPPC manifest interface (schema excerpt, example cards, and signing/logging guidance), see the closed-view auditing note (AnonDHT State 3).[2] The full schema and tooling are distributed out-of-band on request.[8]

7 Limitations and open problems

These recipes ignore correlation structure, adaptivity, and adversarial selection effects; they are intended for sanity checks, not tight privacy accounting. Where correlations matter (e.g. retries in the Boss Fight), prefer mechanism-specific bounds and then convert to endpoint budgets via [3].

8 Takeaways

- **Quadratic amplification:** persistent per-epoch witness gaps (Δ) translate to horizons (T) on the order of $\log_2(1/\eta)/\Delta^2$ (Recipe 1).
- **Cadence is a privacy knob:** the same T epochs can mean hours or months depending on sampling rate; partial visibility scales horizons by $1/q$ (Recipe 2).

- **Alarms cost budget:** tightening false-alarm probability pushes a nontrivial alert-oracle lower bound (Recipe 3); plan to spend privacy budget on operational signals, not just payloads.
- **Don't re-derive endpoint sizing:** convert mechanism bounds to endpoint identification/shortlisting budgets via the endpoint bridge note (Recipe 4).

References

- [1] Anonymous. State-Dependent Anonymity in Anonymous DHTs: Witnesses, Amplification, and Stability. 2026. AnonDHT State series Paper 1 (this archive).
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- [3] Anonymous. Endpoint Metrics Bridge for Anonymous Overlays: Odds Inflation, PML, and MaxL. 2026. Synthesis series note (Synthesis 5) in this archive.
- [4] Anonymous. Trace Interfaces for Anonymous DHTs: Observation Tiers, Committee-Contact Traces, and Exposure Normal Forms. 2026. Synthesis series note (Synthesis 3) in this archive.
- [5] Anonymous. Threat Models and Repetition Windows for Anonymous DHTs. 2026. Synthesis series note (Synthesis 4) in this archive.
- [6] Xinmu Alexis Cao and others. Nym Cryptoeconomic Paper (reward sharing and epochs). 2021. Public Nym document; used only as an example of an epoch/cadence vocabulary.
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- [8] Anonymous. Artifact policy (source-only archive; artifacts available on request). Series note, 2026.
- [9] Anonymous. Notation and Minimal Model for the One-True-Archive. Series synthesis note (Synthesis 8), 2026. (This archive.)